Huawei Access Node (JAY-ACC-77) QoS / Congestion Event Resolution SOP

Node: JAY-ACC-77 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_5614

Description

This Standard Operating Procedure (SOP) addresses QoS / Congestion Events, specifically Microbursts, interface tail drops, and QoS queue starvation, on Huawei Access Nodes. The Huawei Access Node (JAY-ACC-77) is a critical component of the Jayanagar Metro Site, providing access connectivity to numerous subscribers. The QoS / Congestion Event (Alarm Code: ALARM_5614, Severity: Major) is a Major alarm that indicates a potential service impacting condition. The root cause of this failure domain lies in the Huawei Access Node's hardware and software architecture. The Huawei Access Node uses a complex scheduling algorithm to manage QoS queues, which can lead to queue starvation under heavy traffic conditions. Moreover, the Huawei Access Node's interface hardware can experience microbursts, leading to tail drops and further exacerbating congestion. To mitigate these issues, the Huawei Access Node employs various congestion avoidance mechanisms, including Random Early Detection (RED) and Weighted Random Early Detection (WRED). However, under extreme traffic conditions, these mechanisms can be overwhelmed, leading to QoS / Congestion Events. Therefore, it is essential to diagnose and resolve these events promptly to prevent service disruptions and ensure optimal network performance.

Pre-requisite Checks

- Pre-requisite: Verify that the Jayanagar Metro Site is accessible and that the remote hands are dispatched to the site., Description: Ensure that the site is accessible and that the necessary personnel are on-site to facilitate the resolution process.

- Pre-requisite: Check the redundant link status and ensure that it is operational., Description: Verify that the redundant link is up and operational to prevent service disruptions during the resolution process.

Potential Root Causes

Cause	Probability
Queue starvation due to heavy traffic conditions.	High
Microbursts on the interface hardware.	Medium
Insufficient QoS configuration.	Low

Diagnostic Commands & Expected Outputs

```
Command: display queueing queue {interface-name} | include QoS|queue, Expected Output: QoS queue status, Description: This command displays the QoS queue status, including the queue ID, priority, and packet drop count. It helps identify if the queue is experiencing starvation
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or if the packet drop count is high, indicating tail drops.

Command: display interface {interface-name} | include congestion,
Expected Output: Congestion status, Description: This command displays
the congestion status of the interface, including the packet drop
count and the number of packets that have been dropped due to
congestion. It helps identify if the interface is experiencing
microbursts or tail drops.
```

Resolution Steps

Step 1: Step: Modify the QoS configuration to increase the priority of the affected queue., Command: queue-service {queue-id} priority {new-priority}, Description: Modify the QoS configuration to increase the priority of the affected queue to prevent queue starvation. Verify that the priority change is propagated to all members of the queue.

Step 2: Step: Clear the statistics of the affected queue., Command: clear queue-service {queue-id} statistics, Description: Clear the statistics of the affected queue to prevent the packet drop count from accumulating and masking the resolution of the issue.

Step 3: Step: Verify the interface congestion status., Command: display interface {interface-name} | include congestion, Description: Verify that the interface congestion status has improved and that the packet drop count has decreased.

Post-Check Verification

```
Command: display queueing queue {interface-name} | include QoS|queue,
Expected Output: QoS queue status, Description: Verify that the QoS
queue status has improved and that the packet drop count is low.

Command: display interface {interface-name} | include congestion,
Expected Output: Congestion status, Description: Verify that the
interface congestion status has improved and that the packet drop
count is low.
```

Roll-back Procedures

- Step: Restore the original QoS configuration., Command: queue-service {queue-id} priority {original-priority}, Description: Restore the original QoS configuration to prevent unintended consequences.

- Step: Clear the statistics of the affected queue., Command: clear queue-service {queue-id} statistics, Description: Clear the statistics of the affected queue to prevent the packet drop count from accumulating and masking the resolution of the issue.